Chapter 5, Section 3

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1. Let X be a nonsingular projective variety of any dimension, let Y be a nonsingular subvariety, and let $\pi : \tilde{X} \to X$ be obtained by blowing up Y. Show that $p_a(\tilde{X}) = p_a(X)$.

Proof. The proof of (3.4) follows through for the more general case by (II, 8.24).

Proposition 0.1. *In the above setting, we have $\pi_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$, and $\mathrm{R}^i\pi_*\mathcal{O}_{\tilde{X}} = 0$ for $i > 0$. Therefore, $\mathrm{H}^i(X, \mathcal{O}_X) \cong \mathrm{H}^i(\tilde{X}, \mathcal{O}_{\tilde{X}})$ for all $i \geq 0$.*

□

5. Let $a_1, \ldots, a_r$, $r \geq 5$, be distinct elements of k, and let C be the curve in $\mathbb{P}^2$ given by the (affine) equation $y^2 = \prod_{i=1}^r (x - a_i)$. Show that the point P at infinity on the y-axis is a singular point. Compute δ_P and $g(\tilde{Y})$, where $\tilde{Y}$ is the normalization of Y. Show in this way that one obtains hyperelliptic curves of genus $g \geq 2$.

Proof. The point P at infinity on the y-axis has homogenous coordinates $(0 : 1 : 0)$, and the local equation of Y at P at the affine patch defined by $y \neq 0$ is given by $z^{r-2} = \prod_{i=1}^r (x - a_i z)$, so Y has multiplicity $\mu_P(Y) = r - 2$. Let us see what happens when we blow up $\mathbb{P}^2$ at P. The substitution $z = ux$ gives

$$x^{r-2}\left(u^{r-2} - x^2 \prod_{i=1}^r (1 - a_i u)\right) = 0,$$

so the strict transform Y_1 of Y is

$$u^{r-2} - x^2 \prod_{i=1}^r (1 - a_i u) = 0.$$

Since Y_1 is still singular at $(0, 0)$, we blow up Y_1 by making the substitution $x = vu$, which gives

$$u^2\left(u^{r-4} - v^2 \prod_{i=1}^r (1 - a_i u)\right) = 0,$$

so the strict transform Y_2 of Y_1 is

$$u^{r-4} - v^2 \prod_{i=1}^r (1 - a_i u) = 0.$$

We can continue this process until the exponent of u is either 0 or 1, which depends on whether r is even or odd. In the former case, r is even, and the equation of the strict transform of Y after $r/2$ blow-ups is given by

$$1 - w^2 \prod_{i=1}^r (1 - a_i u) = 0,$$

which is nonsingular. The multiplicities at the points that were blown up were $r - 2, 2, \ldots, 2$, so the genus of $\tilde{Y}$ can be computed using (3.9.2)

$$g(\tilde{Y}) = \frac{(r-1)(r-2)}{2} - \frac{(r-2)(r-3)}{2} - \frac{r-4}{2} = \boxed{\frac{r-2}{2}}.$$

If r is odd, then the strict transform of Y after $(r-1)/2$ blow-ups is given by

$$u - w^2 \prod_{i=1}^{r} (1 - a_i u) = 0,$$

which is also nonsingular. Similar to the case r is even, the genus of $\tilde{Y}$ can be computed using (3.9.2)

$$g(\widetilde{Y}) = \frac{(r-1)(r-2)}{2} - \frac{(r-2)(r-3)}{2} - \frac{r-3}{2} = \boxed{\frac{r-1}{2}}.$$

Composing the projection of $Y \to \mathbb{P}^1$ onto the x-axis with the normalization $\tilde{Y} \to Y$ is a g^1_2 on $\tilde{Y}$, so $\tilde{Y}$ is a hyperelliptic curve. □

7. For each of the following singularities at $(0,0)$ in the plane, give an embedded resolution, compute δ_P, and decide which ones are equivalent.

(a) $x^3 + y^5 = 0$
(b) $x^3 + x^4 + y^5 = 0$
(c) $x^3 + y^4 + y^5 = 0$
(d) $x^3 + y^5 + y^6 = 0$
(e) $x^3 + xy^3 + y^5 = 0$

Proof.

(a) The substitution $x = uy$ gives

$$y^3(u^3 + y^2) = 0,$$

so after the first blow-up, we obtain a cuspidal curve. Thus,

$$E_1 : y = 0, \quad C_1 : u^3 + y^2 = 0,$$

where E_1 is the exceptional locus and C_1 is the strict transform of C. Next, the substitution $y = uv$ gives

$$u^2(u + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v = 0.$$

Finally, the substitution $v = wt$ gives

$$w(t + 1) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : t + 1 = 0.$$

The total transform of the original curve is

$$w^{15} t^9 (t + 1) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 4.$$

(b) The substitution $x = uy$ gives

$$y^3(u^3 + u^4y + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + u^4y + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + u^3v + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 = u + u^3v + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v^3w^3 + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v^3w^3 + v = 0.$$

Finally, the substitution $v = wt$ gives

$$w(1 + w^5t^3 + t) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : 1 + w^5t^3 + t = 0.$$

The total transform of the original curve is

$$w^{15}t^9(1 + w^5t^3 + t) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 4.$$

(c) The substitution $x = uy$ gives

$$y^3(u^3 + y + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + y + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u(u^2 + v + uv^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u^2 + v + uv^2 = 0.$$

The substitution $v = uw$ then gives

$$u(u + w + u^2w^2) = 0,$$

so the third blow-up gives

$$E_3 : u = 0, \quad C_3 : u + w + u^2w^2 = 0.$$

Finally, the substitution $u = wt$ gives

$$w(t + 1 + w^3t^2) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : t + 1 + w^3t^2 = 0.$$

The total transform of the original curve is

$$w^{12}t^8(t + 1 + w^3t^2) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 3.$$

(d) The substitution $x = uy$ gives

$$y^3(u^3 + y^2 + y^3) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + y^2 + y^3 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + v^2 + uv^3) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v^2 + uv^3 = 0.$$

The substitution $u = vw$ then gives

$$v(w + v + v^3w) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + v + v^3w = 0.$$

Finally, the substitution $v = wt$ gives

$$w(1 + t + w^3t^3) = 0,$$

so the fourth blow-up gives

$$E_4 : w = 0, \quad C_4 : 1 + t + w^3t^3 = 0.$$

The total transform of the original curve is

$$w^{15}t^9(1 + t + w^3t^3) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) + \frac{1}{2}1(1-1) = 4.$$

(e) The substitution $x = uy$ gives

$$y^3(u^3 + uy + y^2) = 0,$$

so the first blow-up gives

$$E_1 : y = 0, \quad C_1 : u^3 + uy + y^2 = 0.$$

Next, the substitution $y = uv$ gives

$$u^2(u + v + v^2) = 0,$$

so the second blow-up gives

$$E_2 : u = 0, \quad C_2 : u + v + v^2 = 0.$$

The substitution $u = vw$ then gives

$$v(w + 1 + v) = 0,$$

so the third blow-up gives

$$E_3 : v = 0, \quad C_3 : w + 1 + v = 0.$$

The total transform of the original curve is

$$v^9w^5(vw)^2(v^2w)^3(w + 1 + v) = 0,$$

and we compute δ_P using (3.9.3):

$$\delta_P = \frac{1}{2}3(3-1) + \frac{1}{2}2(2-1) + \frac{1}{2}1(1-1) = 4.$$

The singularities in (a), (b), and (d) are equivalent, while the singularities in (c) and (e) are not equivalent to any of the others. □